

The second B200 system is configured.

This system is on Ubuntu 24 LTS.

Current info:

██████████ / ██████████-b200

Updated system chassis fw, uefi fw, gpu adapter fw
configured storage :

/data1 (1.7T)

/data2 (20T)

I'm still working out some cuda init issues with dcgmi diagnostics, but the team can test if they are able to run any or some workloads given this cuda init / dcgmi diagnostics issue.

B200 GPUs are operational under nvidia-driver-580-open. NVLink topology validated

via nvidia-smi. Fabric Manager installation is not supported cleanly in the current

open-driver Ubuntu 24 packaging configuration via apt without transitioning to a server-aligned driver stack.

B200 GPU enumeration is only stable under the open kernel driver path on Ubuntu 24; the server

driver stack installs but does not bind devices correctly on this hardware/kernel combination.

The current trajectory for ongoing support/development is towards the open kernel modules

and decoupled user space tooling.

If requiring Fabric Manager validation, then the system would need to remain on Ubuntu22

with existing server-aligned driver stack, but community supported Ubuntu (22) LTS is until 2027

without extended enterprise maintenance support contracts. Or to wait for Nvidia to correct the

dependencies for the open-branch / fabric manager drivers.

I cross checked with ██████████ / ██████████-b200 regarding the Fabric Manager, and it's currently on

driver 570.86.1, CUDA Version: 12.8, but with the 580-server family drivers:

nvidia-fabricmanager-580,

nvidia-firmware-580-server-580.159.03, nvidia-kernel-common-580-server.

I don't recommend the backward trajectory of the OS or the driver stack implementation for current

term compatibility with newer generations of GPU's available and future generations on the horizon.

==

██████████ 2:43 PM //management

Great! Please try on Monday to see how well it works

==

██████████-b200:~\$

Thu Jun 18 21:21:41 2026

```

+-----+
+-----+
| NVIDIA-SMI 580.167.08                Driver Version: 580.167.08      CUDA Version:
13.0 |
+-----+

```

```

+-----+
| GPU Name                               Persistence-M | Bus-Id        Disp.A | Volatile
Uncorr. ECC |                               Pwr:Usage/Cap |      Memory-Usage | GPU-Util
| Fan  Temp   Perf              |                                     |                   |
Compute M. |                                     |                   |
|                                     |                                     |                   |
MIG M. |                                     |                                     |
|                                     |                                     |                   |
+-----+

```

```

=====|
|  0  NVIDIA B200                      On | 00000000:18:00.0 Off |
0 |
| N/A   39C    P0                      217W / 1000W |      0MiB / 183359MiB |      0%
Default |
|                                     |                                     |
Disabled |
+-----+

```

```

+-----+
|  1  NVIDIA B200                      On | 00000000:19:00.0 Off |
0 |
| N/A   43C    P0                      224W / 1000W |      0MiB / 183359MiB |      0%
Default |
|                                     |                                     |
Disabled |
+-----+

```

```

+-----+
|  2  NVIDIA B200                      On | 00000000:1A:00.0 Off |
0 |
| N/A   40C    P0                      211W / 1000W |      0MiB / 183359MiB |      0%
Default |
|                                     |                                     |
Disabled |
+-----+

```

```

+-----+
|  3  NVIDIA B200                      On | 00000000:1B:00.0 Off |
0 |
| N/A   42C    P0                      211W / 1000W |      0MiB / 183359MiB |      0%
Default |
|                                     |                                     |
Disabled |
+-----+

```

```

+-----+
|  4  NVIDIA B200                      On | 00000000:1C:00.0 Off |
0 |
| N/A   38C    P0                      207W / 1000W |      0MiB / 183359MiB |      0%
Default |
|                                     |                                     |
Disabled |
+-----+

```

```

+-----+
|  5  NVIDIA B200                      On | 00000000:1D:00.0 Off |
0 |
| N/A   41C    P0                      211W / 1000W |      0MiB / 183359MiB |      0%
Default |

```

```

|
Disabled |
+-----+
+-----+
| 6 NVIDIA B200 On | 00000000:BA:00.0 Off |
0 |
| N/A 39C P0 210W / 1000W | 0MiB / 183359MiB | 0%
Default |
|
Disabled |
+-----+
+-----+
| 7 NVIDIA B200 On | 00000000:CA:00.0 Off |
0 |
| N/A 43C P0 212W / 1000W | 0MiB / 183359MiB | 0%
Default |
|
Disabled |
+-----+
+-----+

```

```

+-----+
-----+
| Processes:
| GPU GI CI PID Type Process name GPU
Memory |
| ID ID
Usage |
|
=====

```

```

| No running processes found
|
+-----+

```

```

-----+
[REDACTED]-b200:~$

```

```

[REDACTED]b200:~$
nvcc: NVIDIA (R) Cuda compiler driver
Copyright (c) 2005-2026 NVIDIA Corporation
Built on Fri Apr 24 07:22:02 PM PDT 2026
Cuda compilation tools, release 13.3, V13.3.33
Build cuda_13.3.r13.3/compiler.37862127_0
[REDACTED]-b200:~$

```

```

[REDACTED]-b200:~$
8 GPUs found (Active).

```

```

+-----+
| GPU ID | Device Information |
+-----+
| 0 | Name: NVIDIA B200 |
| | PCI Bus ID: 00000000:18:00.0 |
| | Device UUID: GPU-43eedc98-20de-b0e6-1df7-17ffab96c828 |
+-----+

```

1	Name: NVIDIA B200 PCI Bus ID: 00000000:29:00.0 Device UUID: GPU-2e3a54ca-639f-60a4-1876-871962848ecb
2	Name: NVIDIA B200 PCI Bus ID: 00000000:3A:00.0 Device UUID: GPU-c81cbe41-f9de-896d-4ea5-c1ed39b6693b
3	Name: NVIDIA B200 PCI Bus ID: 00000000:4B:00.0 Device UUID: GPU-37af8fc2-b6e9-ea61-cc6d-2d08a300d299
4	Name: NVIDIA B200 PCI Bus ID: 00000000:9A:00.0 Device UUID: GPU-d25382a0-a726-8acc-469f-235cbb17bbdb
5	Name: NVIDIA B200 PCI Bus ID: 00000000:AA:00.0 Device UUID: GPU-86445f51-f5e1-ed49-1999-ec4b7153e9e9
6	Name: NVIDIA B200 PCI Bus ID: 00000000:BA:00.0 Device UUID: GPU-e367d87a-7522-2404-5593-f882f8077dad
7	Name: NVIDIA B200 PCI Bus ID: 00000000:CA:00.0 Device UUID: GPU-56a9958e-e690-cc5b-0f5b-f4bb3779b6ce

0 NVSwitches found.

Switch ID

ConnectX

0 ConnectX found.

ConnectX

CPU

0 CPUs found.

CPU ID	Device Information

██████████-b200:~\$

██████████-b200:~\$

GPU 0: NVIDIA B200 (UUID: GPU-43eedc98-20de-b0e6-1df7-17ffab96c828)

Link 0: 50 GB/s
Link 1: 50 GB/s
Link 2: 50 GB/s
Link 3: 50 GB/s
Link 4: 50 GB/s
Link 5: 50 GB/s
Link 6: 50 GB/s
Link 7: 50 GB/s
Link 8: 50 GB/s
Link 9: 50 GB/s

Link 10: 50 GB/s
Link 11: 50 GB/s
Link 12: 50 GB/s
Link 13: 50 GB/s
Link 14: 50 GB/s
Link 15: 50 GB/s
Link 16: 50 GB/s
Link 17: 50 GB/s

GPU 1: NVIDIA B200 (UUID: GPU-2e3a54ca-639f-60a4-1876-871962848ecb)

Link 0: 50 GB/s
Link 1: 50 GB/s
Link 2: 50 GB/s
Link 3: 50 GB/s
Link 4: 50 GB/s
Link 5: 50 GB/s
Link 6: 50 GB/s
Link 7: 50 GB/s
Link 8: 50 GB/s
Link 9: 50 GB/s
Link 10: 50 GB/s
Link 11: 50 GB/s
Link 12: 50 GB/s
Link 13: 50 GB/s
Link 14: 50 GB/s
Link 15: 50 GB/s
Link 16: 50 GB/s
Link 17: 50 GB/s

GPU 2: NVIDIA B200 (UUID: GPU-c81cbe41-f9de-896d-4ea5-c1ed39b6693b)

Link 0: 50 GB/s
Link 1: 50 GB/s
Link 2: 50 GB/s
Link 3: 50 GB/s
Link 4: 50 GB/s
Link 5: 50 GB/s
Link 6: 50 GB/s
Link 7: 50 GB/s
Link 8: 50 GB/s
Link 9: 50 GB/s
Link 10: 50 GB/s
Link 11: 50 GB/s
Link 12: 50 GB/s
Link 13: 50 GB/s
Link 14: 50 GB/s
Link 15: 50 GB/s
Link 16: 50 GB/s
Link 17: 50 GB/s

GPU 3: NVIDIA B200 (UUID: GPU-37af8fc2-b6e9-ea61-cc6d-2d08a300d299)

Link 0: 50 GB/s
Link 1: 50 GB/s
Link 2: 50 GB/s
Link 3: 50 GB/s
Link 4: 50 GB/s
Link 5: 50 GB/s
Link 6: 50 GB/s
Link 7: 50 GB/s
Link 8: 50 GB/s
Link 9: 50 GB/s
Link 10: 50 GB/s
Link 11: 50 GB/s

Link 12: 50 GB/s
Link 13: 50 GB/s
Link 14: 50 GB/s
Link 15: 50 GB/s
Link 16: 50 GB/s
Link 17: 50 GB/s

GPU 4: NVIDIA B200 (UUID: GPU-d25382a0-a726-8acc-469f-235cbb17bbdb)

Link 0: 50 GB/s
Link 1: 50 GB/s
Link 2: 50 GB/s
Link 3: 50 GB/s
Link 4: 50 GB/s
Link 5: 50 GB/s
Link 6: 50 GB/s
Link 7: 50 GB/s
Link 8: 50 GB/s
Link 9: 50 GB/s
Link 10: 50 GB/s
Link 11: 50 GB/s
Link 12: 50 GB/s
Link 13: 50 GB/s
Link 14: 50 GB/s
Link 15: 50 GB/s
Link 16: 50 GB/s
Link 17: 50 GB/s

GPU 5: NVIDIA B200 (UUID: GPU-86445f51-f5e1-ed49-1999-ec4b7153e0e9)

Link 0: 50 GB/s
Link 1: 50 GB/s
Link 2: 50 GB/s
Link 3: 50 GB/s
Link 4: 50 GB/s
Link 5: 50 GB/s
Link 6: 50 GB/s
Link 7: 50 GB/s
Link 8: 50 GB/s
Link 9: 50 GB/s
Link 10: 50 GB/s
Link 11: 50 GB/s
Link 12: 50 GB/s
Link 13: 50 GB/s
Link 14: 50 GB/s
Link 15: 50 GB/s
Link 16: 50 GB/s
Link 17: 50 GB/s

GPU 6: NVIDIA B200 (UUID: GPU-e367d87a-7522-2404-5593-f882f8077dad)

Link 0: 50 GB/s
Link 1: 50 GB/s
Link 2: 50 GB/s
Link 3: 50 GB/s
Link 4: 50 GB/s
Link 5: 50 GB/s
Link 6: 50 GB/s
Link 7: 50 GB/s
Link 8: 50 GB/s
Link 9: 50 GB/s
Link 10: 50 GB/s
Link 11: 50 GB/s
Link 12: 50 GB/s
Link 13: 50 GB/s

```

Link 14: 50 GB/s
Link 15: 50 GB/s
Link 16: 50 GB/s
Link 17: 50 GB/s
GPU 7: NVIDIA B200 (UUID: GPU-56a9958e-e690-cc5b-0f5b-f4bb3779b6ce)
Link 0: 50 GB/s
Link 1: 50 GB/s
Link 2: 50 GB/s
Link 3: 50 GB/s
Link 4: 50 GB/s
Link 5: 50 GB/s
Link 6: 50 GB/s
Link 7: 50 GB/s
Link 8: 50 GB/s
Link 9: 50 GB/s
Link 10: 50 GB/s
Link 11: 50 GB/s
Link 12: 50 GB/s
Link 13: 50 GB/s
Link 14: 50 GB/s
Link 15: 50 GB/s
Link 16: 50 GB/s
Link 17: 50 GB/s
[REDACTED]-b200:~$

```

```

[REDACTED]-b200:~$
GPU0 GPU1 GPU2 GPU3 GPU4 GPU5 GPU6 GPU7 CPU
Affinity NUMA Affinity GPU NUMA ID
GPU0 X NV18 NV18 NV18 NV18 NV18 NV18 NV18 0-47,96-143
0 N/A
GPU1 NV18 X NV18 NV18 NV18 NV18 NV18 NV18 0-47,96-143
0 N/A
GPU2 NV18 NV18 X NV18 NV18 NV18 NV18 NV18 0-47,96-143
0 N/A
GPU3 NV18 NV18 NV18 X NV18 NV18 NV18 NV18 0-47,96-143
0 N/A
GPU4 NV18 NV18 NV18 NV18 X NV18 NV18 NV18 48-95,144-
191 1 N/A
GPU5 NV18 NV18 NV18 NV18 NV18 X NV18 NV18 48-95,144-
191 1 N/A
GPU6 NV18 NV18 NV18 NV18 NV18 NV18 X NV18 48-95,144-
191 1 N/A
GPU7 NV18 NV18 NV18 NV18 NV18 NV18 NV18 X 48-95,144-
191 1 N/A

```

Legend:

```

X = Self
SYS = Connection traversing PCIe as well as the SMP interconnect between NUMA
nodes (e.g., QPI/UPI)
NODE = Connection traversing PCIe as well as the interconnect between PCIe Host
Bridges within a NUMA node
PHB = Connection traversing PCIe as well as a PCIe Host Bridge (typically the
CPU)
PXB = Connection traversing multiple PCIe bridges (without traversing the PCIe
Host Bridge)
PIX = Connection traversing at most a single PCIe bridge

```

NV# = Connection traversing a bonded set of # NVLinks

-b200:~\$

-b200:~\$

NAME	USED	AVAIL	REFER	MOUNTPPOINT
data1	540K	1.68T	96K	/data1
data2	1.22M	19.7T	205K	/data2

-b200:~\$

pool: data1
state: ONLINE
config:

NAME	STATE	READ	WRITE	CKSUM
data1	ONLINE	0	0	0
nvme1n1	ONLINE	0	0	0

errors: No known data errors

pool: data2
state: ONLINE
config:

NAME	STATE	READ	WRITE	CKSUM
data2	ONLINE	0	0	0
raidz2-0	ONLINE	0	0	0
nvme2n1	ONLINE	0	0	0
nvme3n1	ONLINE	0	0	0
nvme4n1	ONLINE	0	0	0
nvme5n1	ONLINE	0	0	0
nvme6n1	ONLINE	0	0	0
nvme7n1	ONLINE	0	0	0
nvme8n1	ONLINE	0	0	0
nvme9n1	ONLINE	0	0	0

errors: No known data errors

-b200:~\$

-b200:~\$

Filesystem	Size	Used	Avail	Use%	Mounted on
tmpfs	303G	4.5M	303G	1%	/run
efivarfs	512K	277K	231K	55%	/sys/firmware/efi/efivars
/dev/mapper/ubuntu--vg-ubuntu--lv	98G	22G	72G	24%	/
tmpfs	1.5T	0	1.5T	0%	/dev/shm
tmpfs	5.0M	0	5.0M	0%	/run/lock
/dev/nvme0n1p2	2.0G	201M	1.6G	11%	/boot
/dev/nvme0n1p1	1.1G	6.2M	1.1G	1%	/boot/efi
/dev/mapper/ubuntu--vg-lv--0	1.7T	33G	1.7T	2%	/home
data1	1.7T	128K	1.7T	1%	/data1
data2	20T	256K	20T	1%	/data2
tmpfs	303G	20K	303G	1%	/run/user/1000

-b200:~\$

██████-b200:~\$

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINTS
nvme6n1	259:0	0	3.5T	0	disk	
└nvme6n1p1	259:1	0	3.5T	0	part	
└nvme6n1p9	259:2	0	8M	0	part	
nvme1n1	259:3	0	1.7T	0	disk	
└nvme1n1p1	259:4	0	1.7T	0	part	
└nvme1n1p9	259:5	0	8M	0	part	
nvme5n1	259:6	0	3.5T	0	disk	
└nvme5n1p1	259:16	0	3.5T	0	part	
└nvme5n1p9	259:17	0	8M	0	part	
nvme3n1	259:7	0	3.5T	0	disk	
└nvme3n1p1	259:14	0	3.5T	0	part	
└nvme3n1p9	259:15	0	8M	0	part	
nvme0n1	259:8	0	1.7T	0	disk	
└nvme0n1p1	259:24	0	1G	0	part	/boot/efi
└nvme0n1p2	259:25	0	2G	0	part	/boot
└nvme0n1p3	259:26	0	1.7T	0	part	
└ubuntu--vg-ubuntu--lv	252:0	0	100G	0	lvm	/
└ubuntu--vg-lv--0	252:1	0	1.6T	0	lvm	/home
nvme8n1	259:9	0	3.5T	0	disk	
└nvme8n1p1	259:20	0	3.5T	0	part	
└nvme8n1p9	259:21	0	8M	0	part	
nvme2n1	259:10	0	3.5T	0	disk	
└nvme2n1p1	259:18	0	3.5T	0	part	
└nvme2n1p9	259:19	0	8M	0	part	
nvme7n1	259:11	0	3.5T	0	disk	
└nvme7n1p1	259:22	0	3.5T	0	part	
└nvme7n1p9	259:23	0	8M	0	part	
nvme9n1	259:12	0	3.5T	0	disk	
└nvme9n1p1	259:29	0	3.5T	0	part	
└nvme9n1p9	259:30	0	8M	0	part	
nvme4n1	259:13	0	3.5T	0	disk	
└nvme4n1p1	259:27	0	3.5T	0	part	
└nvme4n1p9	259:28	0	8M	0	part	

██████-b200:~\$

====

// another user 'fixed' the fabric manager, but it was only for local in memory

post reboot after ██████ fixed fabric manager with the default install
removing the 580 open drivers, and the server branch installed

fabric manager was working in the current running instance/shell before reboot

#no output, no mention of modules loaded

██████-b200:~\$ sudo dmesg | grep -i nvidia

██████200:~\$ sudo dmesg | grep -i gpu

██████-b200:~\$ nvidia-smi

NVIDIA-SMI has failed because it couldn't communicate with the NVIDIA driver. Make sure that the latest NVIDIA driver is installed and running.

```

-b200:~$
0 GPUs found (Active).
+-----+
| GPU ID | Device Information |
+-----+
+-----+
0 NvSwitches found.
+-----+
| Switch ID |
+-----+
+-----+
0 ConnectX found.
+-----+
| ConnectX |
+-----+
+-----+
0 CPUs found.
+-----+
| CPU ID | Device Information |
+-----+
+-----+
-b200:~$
```

#no output, no mention of modules loaded

```

-b200:~$ cat /proc/modules | grep -i nvidia
-b200:~$
```

#no output, no mention of modules loaded

```

-b200:~$ ls -l /proc/driver/nvidia
ls: cannot access '/proc/driver/nvidia': No such file or directory
-b200:~$
```

#hardware is recognized, but not meaningfully initialized for functionality

```

-b200:~$ lspci | grep -i nvidia
18:00.0 3D controller: NVIDIA Corporation Device 2901 (rev a1)
29:00.0 3D controller: NVIDIA Corporation Device 2901 (rev a1)
3a:00.0 3D controller: NVIDIA Corporation Device 2901 (rev a1)
4b:00.0 3D controller: NVIDIA Corporation Device 2901 (rev a1)
9a:00.0 3D controller: NVIDIA Corporation Device 2901 (rev a1)
aa:00.0 3D controller: NVIDIA Corporation Device 2901 (rev a1)
ba:00.0 3D controller: NVIDIA Corporation Device 2901 (rev a1)
ca:00.0 3D controller: NVIDIA Corporation Device 2901 (rev a1)
-b200
```

#no mention of modules loaded

██████-b200:~\$ sudo modprobe nvidia

modprobe: FATAL: Module nvidia not found in directory /lib/modules/6.8.0-124-generic

██████-b200:~\$

====

reconfigured to verify fabric manager functionality / persistence after reboot on the 580/open driver

██████-b200:~\$

Tue Jun 23 17:35:24 2026

```
+-----+
+-----+
| NVIDIA-SMI 580.167.08                Driver Version: 580.167.08      CUDA Version:
13.0   |
+-----+-----+-----+-----+
+-----+
| GPU  Name                               Persistence-M | Bus-Id               Disp.A | Volatile
Uncorr. ECC |                               Pwr:Usage/Cap |      Memory-Usage | GPU-Util
| Fan  Temp    Perf                Pwr:Usage/Cap |      Memory-Usage | GPU-Util
Compute M. |                               |                      |
| MIG M. |                               |                      |
+-----+-----+-----+-----+
=====
|   0   NVIDIA B200                      On | 00000000:18:00.0 Off |
0 |
| N/A   39C    P0                211W / 1000W |      0MiB / 183359MiB |      0%
Default |
| Disabled |
+-----+-----+-----+-----+
+-----+
|   1   NVIDIA B200                      On | 00000000:29:00.0 Off |
0 |
| N/A   43C    P0                217W / 1000W |      0MiB / 183359MiB |      0%
Default |
| Disabled |
+-----+-----+-----+-----+
+-----+
|   2   NVIDIA B200                      On | 00000000:3A:00.0 Off |
0 |
| N/A   40C    P0                211W / 1000W |      0MiB / 183359MiB |      0%
Default |
| Disabled |
+-----+-----+-----+-----+
+-----+
|   3   NVIDIA B200                      On | 00000000:4B:00.0 Off |
0 |
| N/A   42C    P0                207W / 1000W |      0MiB / 183359MiB |      0%
```

```

Default |
|
Disabled |
+-----+-----+-----+
| 4 NVIDIA B200 On | 00000000:9A:00.0 Off |
| 0 |
| N/A 38C P0 207W / 1000W | 0MiB / 183359MiB | 0%
Default |
|
Disabled |
+-----+-----+-----+
| 5 NVIDIA B200 On | 00000000:AA:00.0 Off |
| 0 |
| N/A 42C P0 211W / 1000W | 0MiB / 183359MiB | 0%
Default |
|
Disabled |
+-----+-----+-----+
| 6 NVIDIA B200 On | 00000000:BA:00.0 Off |
| 0 |
| N/A 39C P0 210W / 1000W | 0MiB / 183359MiB | 0%
Default |
|
Disabled |
+-----+-----+-----+
| 7 NVIDIA B200 On | 00000000:CA:00.0 Off |
| 0 |
| N/A 43C P0 212W / 1000W | 0MiB / 183359MiB | 0%
Default |
|
Disabled |
+-----+-----+-----+

```

```

+-----+
| Processes:
| GPU GI CI PID Type Process name
GPU Memory |
| ID ID
Usage |
|=====
=====|
| No running processes found
|
+-----+

```

```

-b200:~$
-b200:~$ nvcc --version
nvcc: NVIDIA (R) Cuda compiler driver
Copyright (c) 2005-2026 NVIDIA Corporation
Built on Fri_Apr_24_07:22:02_PM_PDT_2026
Cuda compilation tools, release 13.3, V13.3.33

```

Build cuda_13.3.r13.3/compiler.37862127_0

██████-b200:~\$

██████-b200:~\$

nvidia/ nvidia-caps/ nvidia-modeset/ nvidia-nvlink/
nvidia-nvswitch/ nvidia-uvm/

██████-b200:~\$

██████-b200:~\$

nvidia_uvm	2158592	4
nvidia_drm	139264	0
nvidia_modeset	1822720	1 nvidia_drm
nvidia	14417920	73 nvidia_uvm,nvidia_modeset
video	77824	1 nvidia_modeset
ecc	45056	1 nvidia

██████-b200:~\$

██████-b200:~\$

8 GPUs found (Active).

GPU ID Device Information	
0	Name: NVIDIA B200 PCI Bus ID: 00000000:18:00.0 Device UUID: GPU-43eedc98-20de-b0e6-1df7-17ffab96c828
1	Name: NVIDIA B200 PCI Bus ID: 00000000:29:00.0 Device UUID: GPU-2e3a54ca-639f-60a4-1876-871962848ecb
2	Name: NVIDIA B200 PCI Bus ID: 00000000:3A:00.0 Device UUID: GPU-c81cbe41-f9de-896d-4ea5-c1ed39b6693b
3	Name: NVIDIA B200 PCI Bus ID: 00000000:4B:00.0


```

-b200:~$
18:00.0 3D controller: NVIDIA Corporation Device 2901 (rev a1)
29:00.0 3D controller: NVIDIA Corporation Device 2901 (rev a1)
3a:00.0 3D controller: NVIDIA Corporation Device 2901 (rev a1)
4b:00.0 3D controller: NVIDIA Corporation Device 2901 (rev a1)
9a:00.0 3D controller: NVIDIA Corporation Device 2901 (rev a1)
aa:00.0 3D controller: NVIDIA Corporation Device 2901 (rev a1)
ba:00.0 3D controller: NVIDIA Corporation Device 2901 (rev a1)
ca:00.0 3D controller: NVIDIA Corporation Device 2901 (rev a1)
-b200:~$

```

```

-b200:~$
      GPU0      GPU1      GPU2      GPU3      GPU4      GPU5      GPU6      GPU7 CPU Affinity
NUMA Affinity  GPU NUMA ID
GPU0      X      NV18      NV18      NV18      NV18      NV18      NV18 0-47,96-143
0
GPU1      NV18      X      NV18      NV18      NV18      NV18      NV18 0-47,96-143
0
GPU2      NV18      NV18      X      NV18      NV18      NV18      NV18 0-47,96-143
0
GPU3      NV18      NV18      NV18      X      NV18      NV18      NV18 0-47,96-143
0
GPU4      NV18      NV18      NV18      NV18      X      NV18      NV18 48-95,144-191
1
GPU5      NV18      NV18      NV18      NV18      NV18      X      NV18 48-95,144-191
1
GPU6      NV18      NV18      NV18      NV18      NV18      X      NV18 48-95,144-191
1
GPU7      NV18      NV18      NV18      NV18      NV18      NV18      X 48-95,144-191
1

```

Legend:

```

X      = Self
SYS    = Connection traversing PCIe as well as the SMP interconnect between NUMA
nodes (e.g., QPI/UPI)
NODE   = Connection traversing PCIe as well as the interconnect between PCIe Host
Bridges within a NUMA node
PHB    = Connection traversing PCIe as well as a PCIe Host Bridge (typically the
CPU)
PXB    = Connection traversing multiple PCIe bridges (without traversing the PCIe
Host Bridge)
PIX    = Connection traversing at most a single PCIe bridge
NV#    = Connection traversing a bonded set of # NVLinks

```

```

-b200:~$
-b200:~$
GPU 0: NVIDIA B200 (UUID: GPU-43eedc98-20de-b0e6-1df7-17ffab96c828)
GPU 1: NVIDIA B200 (UUID: GPU-2e3a54ca-639f-60a4-1876-871962848ecb)
GPU 2: NVIDIA B200 (UUID: GPU-c81cbe41-f9de-896d-4ea5-c1ed39b6693b)
GPU 3: NVIDIA B200 (UUID: GPU-37af8fc2-b6e9-ea61-cc6d-2d08a300d299)
GPU 4: NVIDIA B200 (UUID: GPU-d25382a0-a726-8acc-469f-235cbb17bbdb)
GPU 5: NVIDIA B200 (UUID: GPU-86445f51-f5e1-ed49-1999-ec4b7153e0e9)

```

```
GPU 6: NVIDIA B200 (UUID: GPU-e367d87a-7522-2404-5593-f882f8077dad)
GPU 7: NVIDIA B200 (UUID: GPU-56a9958e-e690-cc5b-0f5b-f4bb3779b6ce)
[REDACTED]-b200:~$
```

```
[REDACTED]-b200:~$
• nvidia-fabricmanager.service - NVIDIA fabric manager service
  Loaded: loaded (/usr/lib/systemd/system/nvidia-fabricmanager.service;
  enabled; preset: enabled)
  Active: active (running) since Wed 2026-06-24 07:55:17 UTC; 1min 54s ago
  Process: 6311 ExecStartPre=/usr/bin/nvidia-fabricmanager-start.sh --mode
  precheck (code=exited, status=0/SUCCESS)
  Process: 6332 ExecStart=/usr/bin/nvidia-fabricmanager-start.sh --mode start
  (code=exited, status=0/SUCCESS)
  Main PID: 6461 (nv-fabricmanager)
  Tasks: 90 (limit: 629145)
  Memory: 122.7M (peak: 123.3M)
  CPU: 4.461s
  CGroup: /system.slice/nvidia-fabricmanager.service
          └─6436 /opt/nvidia/nvlsms/sbin/nvlsms -F
/usr/share/nvidia/nvlsms/nvlsms.conf -B --pid_file /var/run/nvidia-fabricman>
          └─6439 osm_crashd
          └─6461 /usr/bin/nv-fabricmanager -c
/usr/share/nvidia/nvswitch/fabricmanager.cfg -g 0xc470bd0300168612
```

```
Jun 24 07:55:09 [REDACTED]-b200 OpenSM[6433]: -I- Configuration loaded
Jun 24 07:55:09 [REDACTED]-b200 nvidia-fabricmanager-start.sh[6332]: Started
"Nvidia NVLink Subnet Manager"
Jun 24 07:55:09 [REDACTED]-b200 OpenSM[6436]: /var/log/nvlsms.log log file opened
Jun 24 07:55:09 [REDACTED]-b200 OpenSM[6436]: OpenSM
2025.10.14_42349a2_821fcee_ae214d2
Jun 24 07:55:09 [REDACTED]-b200 OpenSM[6436]: Entering DISCOVERING state
Jun 24 07:55:10 [REDACTED]-b200 OpenSM[6436]: Entering MASTER state
Jun 24 07:55:17 [REDACTED]-b200 nv-fabricmanager[6461]: NodeId 0 partition id 57082
is activated.
Jun 24 07:55:17 [REDACTED]-b200 nv-fabricmanager[6461]: Successfully configured all
the available GPUs and NVSwitches to route NV>
Jun 24 07:55:17 [REDACTED]-b200 nvidia-fabricmanager-start.sh[6332]: Started
"Nvidia Fabric Manager"
Jun 24 07:55:17 [REDACTED]-b200 systemd[1]: Started nvidia-fabricmanager.service -
NVIDIA fabric manager service.
[REDACTED]-b200:~$
```

```
[REDACTED]-b200:~$
```

WARNING: apt does not have a stable CLI interface. Use with caution in scripts.

```
libnvidia-cfg1-580/unknown,now 580.167.08-1ubuntu1 amd64 [installed,automatic]
libnvidia-compute-580/unknown,now 580.167.08-1ubuntu1 amd64
[installed,automatic]
libnvidia-decode-580/unknown,now 580.167.08-1ubuntu1 amd64 [installed,automatic]
```



```

libnvidia-egl-gbm1/unknown,now 1.1.3-1ubuntu1 amd64 [installed,automatic]
libnvidia-egl-wayland1/unknown,now 1:1.1.21-1ubuntu1 amd64 [installed,automatic]
libnvidia-egl-xcb1/unknown,now 1.0.5-1ubuntu1 amd64 [installed,automatic]
libnvidia-egl-xlib1/unknown,now 1.0.5-1ubuntu1 amd64 [installed,automatic]
libnvidia-encode-580/unknown,now 580.167.08-1ubuntu1 amd64 [installed,automatic]
libnvidia-fbc1-580/unknown,now 580.167.08-1ubuntu1 amd64 [installed,automatic]
libnvidia-gli-580/unknown,now 580.167.08-1ubuntu1 amd64 [installed,automatic]
libnvidia-gpucomp-580/unknown,now 580.167.08-1ubuntu1 amd64 [installed,automatic]
nvidia-dkms-580-open/unknown,now 580.167.08-1ubuntu1 amd64 [installed,automatic]
nvidia-driver-580-open/unknown,now 580.167.08-1ubuntu1 amd64 [installed]
nvidia-driver-pinning-580/unknown,now 580-1ubuntu3 all [installed]
nvidia-fabricmanager/unknown,now 580.167.08-1ubuntu1 amd64 [installed]
nvidia-firmware-580-server-580.159.03/noble-updates,noble-security,now 580.159.03-
0ubuntu0.24.04.1
amd64 [installed,auto-removable]
nvidia-firmware-580/unknown,now 580.167.08-1ubuntu1 amd64 [installed,automatic]
nvidia-kernel-common-580/unknown,now 580.167.08-1ubuntu1 amd64
[installed,automatic]
nvidia-kernel-source-580-open/unknown,now 580.167.08-1ubuntu1 amd64 [installed]
nvidia-modprobe/unknown,now 610.43.02-1ubuntu1 amd64 [installed,automatic]
nvidia-open/unknown,now 580.167.08-1ubuntu1 amd64 [installed]
nvidia-persistenced/unknown,now 610.43.02-1ubuntu1 amd64 [installed,automatic]
nvidia-settings/unknown,now 610.43.02-1ubuntu1 amd64 [installed,automatic]
nvidia-utils-580/unknown,now 580.167.08-1ubuntu1 amd64 [installed]
xserver-xorg-video-nvidia-580/unknown,now 580.167.08-1ubuntu1 amd64
[installed,automatic]

```

```

-b200:~$

```

```

-b200:~$ sudo systemctl restart nvidia-dcgm.service
-b200:~$

```

```

-b200:~$

```

Successfully ran diagnostic for group.

Diagnostic	Result
----- Metadata -----	
DCGM Version	4.6.0
Driver Version Detected	580.167.08
GPU Device IDs Detected	2901, 2901, 2901, 2901, 2901, 2901, 2901, 2901
----- Deployment -----	
Software	Pass
	GPU0: Pass
	GPU1: Pass
	GPU2: Pass
	GPU3: Pass
	GPU4: Pass
	GPU5: Pass
	GPU6: Pass
	GPU7: Pass

-b200:~\$

```
-b200:~$ nvidia-smi --query-gpu=fabric.state,fabric.status --format=csv
fabric.state, fabric.status
Completed, Success
Completed, Success
Completed, Success
Completed, Success
Completed, Success
Completed, Success
Completed, Success
Completed, Success
Completed, Success
-b200:~$
```

-b200:~\$

Successfully ran diagnostic for group.

Diagnostic		Result
----- Metadata -----		
DCGM Version		4.6.0
Driver Version Detected		580.167.08
GPU Device IDs Detected		2901, 2901, 2901, 2901, 2901, 2901, 2901, 2901
----- Deployment -----		
software		Pass
		GPU0: Pass
		GPU1: Pass
		GPU2: Pass
		GPU3: Pass
		GPU4: Pass
		GPU5: Pass
		GPU6: Pass
		GPU7: Pass
----- Hardware -----		
memory		Pass
		GPU0: Pass
		GPU1: Pass
		GPU2: Pass
		GPU3: Pass
		GPU4: Pass
		GPU5: Pass
		GPU6: Pass
		GPU7: Pass
----- Integration -----		
pcie		Pass
		GPU0: Pass
		GPU1: Pass
		GPU2: Pass
		GPU3: Pass
		GPU4: Pass
		GPU5: Pass
		GPU6: Pass
		GPU7: Pass

-b200:~\$


```
// -h200 proper redeployment
// upgrade os from ubuntu 22 to ubuntu 24
// patch os
// system chassis component uefi/bios/bmc/xcc/gpu fw updates
// gpu os drivers / supplemental apps
// system/gpu diags/validation
```

```
//existing os
```

```
@-h200:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 22.04.5 LTS"
NAME="Ubuntu"
VERSION_ID="22.04"
VERSION="22.04.5 LTS (Jammy Jellyfish)"
VERSION_CODENAME=jammy
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=jammy
-h200:~$
```

```
//existing gpu fw driver ver:
```

NVIDIA	N/A	10DE18BE	96.00.A5.00.03
--------	-----	----------	----------------

17	NVIDIA H200 Display Controller	PCI-E 2335-895-A1
18	NVIDIA H200 Display Controller	PCI-E 2335-895-A1
19	NVIDIA H200 Display Controller	PCI-E 2335-895-A1
20	NVIDIA H200 Display Controller	PCI-E 2335-895-A1
21	NVIDIA H200 Display Controller	PCI-E 2335-895-A1
22	NVIDIA H200 Display Controller	PCI-E 2335-895-A1
23	NVIDIA H200 Display Controller	PCI-E 2335-895-A1
24	NVIDIA H200 Display Controller	PCI-E 2335-895-A1

```
//existing system fw
```

BMC (Primary)	Active	10.10 R3X329G	2025-12-17
BMC (Backup)	Inactive	3.13 DVI399T	2025-03-13
FPGA	Active	02.44 Altera	2024-12-23
UEFI	Active	3.30 R3E111D	2025-11-07

```
// updating
```

Device Name	New Version	Installed Version
UEFI	R3E115C	R3E111D
BMC (Primary)	R3X331E	R3X329G
BMC (Backup)	R3X331E	DVI399T

Slot No.	Device Name	New Version	Installed Version
17	NVIDIA H200	96.00.da.00.11	96.00.A5.00.03
18	NVIDIA H200	96.00.da.00.11	96.00.A5.00.03

19	NVIDIA H200	96.00.da.00.11	96.00.A5.00.03
20	NVIDIA H200	96.00.da.00.11	96.00.A5.00.03
21	NVIDIA H200	96.00.da.00.11	96.00.A5.00.03
22	NVIDIA H200	96.00.da.00.11	96.00.A5.00.03
23	NVIDIA H200	96.00.da.00.11	96.00.A5.00.03
24	NVIDIA H200	96.00.da.00.11	96.00.A5.00.03

```
// update os
```

```
@-h200:~$ cat /etc/os-release
```

```
PRETTY_NAME="Ubuntu 24.04.4 LTS"
```

```
NAME="Ubuntu"
```

```
VERSION_ID="24.04"
```

```
VERSION="24.04.4 LTS (Noble Numbat)"
```

```
VERSION_CODENAME=noble
```

```
ID=ubuntu
```

```
ID_LIKE=debian
```

```
HOME_URL="https://www.ubuntu.com/"
```

```
SUPPORT_URL="https://help.ubuntu.com/"
```

```
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
```

```
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
```

```
UBUNTU_CODENAME=noble
```

```
LOGO=ubuntu-logo
```

```
@-h200:~$
```

```
// upgraded 2x 15tb disks into raid 1 mirror for resiliency
```

```
// verified access permissions, created test file
```

```
@-h200:~$
```

```
pool: r1-data
```

```
state: ONLINE
```

```
config:
```

NAME	STATE	READ	WRITE	CKSUM
r1-data	ONLINE	0	0	0
mirror-0	ONLINE	0	0	0
nvme1n1	ONLINE	0	0	0
nvme2n1	ONLINE	0	0	0

```
errors: No known data errors
```

```
@-h200:~$
```

```
@-h200:~$
```

NAME	USED	AVAIL	REFER	MOUNTPPOINT
r1-data	144K	13.8T	24K	/data

```
@-h200:~$
```

```
@-h200:~$
```

```
total 8.1G
```

```
[truncated]
```

```
512 drwxrwxrwx 2 root root 3 Aug 20 11:04 data
```

```
[truncated]
```

```
@-h200:~$ ls -lsh /data/
```

```
total 512
```

```
512 -rw-rw-r-- 1 0 Aug 20 11:04 test-file
```

```
@-h200:~$
```

```
@-h200:~$
```

Filesystem	Size	Used	Avail	Use%	Mounted on
[truncated]					
/dev/mapper/ubuntu--vg-home	792G	5.8M	790G	1%	/home
tmpfs	202G	20K	202G	1%	/run/user/1000
r1-data	14T	128K	14T	1%	/data

```
@-h200:~$
```

```
//configured network for dhcp persistence on specified ports for netplan/cloud cfg  
(subiquity install)
```

```
// post system drivers gpu validations
```

```
@-h200:~$
```

```
libnvidia-cfg1-570/unknown,now 570.211.01-0ubuntu1 amd64 [installed,automatic]  
libnvidia-common-570/unknown,now 570.211.01-0ubuntu1 all [installed,automatic]  
libnvidia-compute-570/unknown,now 570.211.01-0ubuntu1 amd64 [installed,automatic]  
libnvidia-decode-570/unknown,now 570.211.01-0ubuntu1 amd64 [installed,automatic]  
libnvidia-egl-gbm1/unknown,now 1.1.3-1ubuntu1 amd64 [installed,automatic]  
libnvidia-egl-wayland1/unknown,now 1:1.1.21-1ubuntu1 amd64 [installed,automatic]  
libnvidia-egl-xcb1/unknown,now 1.0.5-1ubuntu1 amd64 [installed,automatic]  
libnvidia-egl-xlib1/unknown,now 1.0.5-1ubuntu1 amd64 [installed,automatic]  
libnvidia-encode-570/unknown,now 570.211.01-0ubuntu1 amd64 [installed,automatic]  
libnvidia-extra-570/unknown,now 570.211.01-0ubuntu1 amd64 [installed,automatic]  
libnvidia-fbc1-570/unknown,now 570.211.01-0ubuntu1 amd64 [installed,automatic]  
libnvidia-gl-570/unknown,now 570.211.01-0ubuntu1 amd64 [installed,automatic]  
libnvidia-gpucomp-580/unknown,now 580.178.04-1ubuntu1 amd64 [installed,auto-  
removable]  
libnvidia-nscq-570/unknown,now 570.211.01-1 amd64 [installed]  
nvidia-dkms-570-open/unknown,now 570.211.01-0ubuntu1 amd64 [installed,automatic]  
nvidia-driver-570-open/unknown,now 570.211.01-0ubuntu1 amd64 [installed,automatic]  
nvidia-fabricmanager-570/unknown,now 570.211.01-1 amd64 [installed]  
nvidia-firmware-570/unknown,now 570.211.01-0ubuntu1 amd64 [installed,automatic]  
nvidia-kernel-common-570/unknown,now 570.211.01-0ubuntu1 amd64  
[installed,automatic]  
nvidia-kernel-source-570-open/unknown,now 570.211.01-0ubuntu1 amd64  
[installed,automatic]  
nvidia-modprobe/unknown,now 610.57.04-1ubuntu1 amd64 [installed,automatic]  
nvidia-open-570/unknown,now 570.211.01-0ubuntu1 amd64 [installed]  
nvidia-persistenced/unknown,now 610.57.04-1ubuntu1 amd64 [installed,automatic]  
nvidia-settings/unknown,now 610.57.04-1ubuntu1 amd64 [installed,automatic]  
xserver-xorg-video-nvidia-570/unknown,now 570.211.01-0ubuntu1 amd64  
[installed,automatic]
```

```
@-h200:~$
```

```
@-h200:~$
```

ii	cuda-nsight-compute-13-3	13.3.1-1
amd64	NVIDIA Nsight Compute	
ii	cuda-nsight-systems-13-3	13.3.1-1
amd64	NVIDIA Nsight Systems	
ii	cuda-nvtx-13-3	13.3.29-1
amd64	NVIDIA Tools Extension	

```

ii datacenter-gpu-manager 1:3.3.9
amd64 NVIDIA® Datacenter GPU Management Tools
ii libnvidia-cfg1-570:amd64 570.211.01-0ubuntu1
amd64 NVIDIA binary OpenGL/GLX configuration library
ii libnvidia-common-570 570.211.01-0ubuntu1
all Shared files used by the NVIDIA libraries
ii libnvidia-compute-570:amd64 570.211.01-0ubuntu1
amd64 NVIDIA libcompute package
ii libnvidia-decode-570:amd64 570.211.01-0ubuntu1
amd64 NVIDIA Video Decoding runtime libraries
ii libnvidia-egl-gbm1:amd64 1.1.3-1ubuntu1
amd64 GBM EGL external platform library for NVIDIA
ii libnvidia-egl-wayland1:amd64 1:1.1.21-1ubuntu1
amd64 Wayland EGL External Platform library -- shared library
ii libnvidia-egl-xcb1:amd64 1.0.5-1ubuntu1
amd64 This is an EGL platform library for the NVIDIA driver to support
ii libnvidia-egl-xlib1:amd64 1.0.5-1ubuntu1
amd64 This is an EGL platform library for the NVIDIA driver to support
ii libnvidia-encode-570:amd64 570.211.01-0ubuntu1
amd64 NVENC Video Encoding runtime library
ii libnvidia-extra-570:amd64 570.211.01-0ubuntu1
amd64 Extra libraries for the NVIDIA driver
ii libnvidia-fbc1-570:amd64 570.211.01-0ubuntu1
amd64 NVIDIA OpenGL-based Framebuffer Capture runtime library
ii libnvidia-gl-570:amd64 570.211.01-0ubuntu1
amd64 NVIDIA OpenGL/GLX/EGL/GLES GLVND libraries and Vulkan ICD
ii libnvidia-gpucomp-580:amd64 580.178.04-1ubuntu1
amd64 NVIDIA binary GPU compiler library
ii libnvidia-nsqc-570 570.211.01-1
amd64 NVSwitch Configuration and Query library
ii nsight-compute-2026.2.1 2026.2.1.5-1
amd64 NVIDIA Nsight Compute
ii nvidia-dkms-570-open 570.211.01-0ubuntu1
amd64 NVIDIA DKMS package (open kernel module)
ii nvidia-driver-570-open 570.211.01-0ubuntu1
amd64 NVIDIA driver (open kernel) metapackage
ii nvidia-fabricmanager-570 570.211.01-1
amd64 Fabric Manager for NVSwitch based systems.
ii nvidia-firmware-570 570.211.01-0ubuntu1
amd64 Firmware files used by the kernel module
ii nvidia-kernel-common-570 570.211.01-0ubuntu1
amd64 Shared files used with the kernel module
ii nvidia-kernel-source-570-open 570.211.01-0ubuntu1
amd64 NVIDIA kernel source package
ii nvidia-modprobe 610.57.04-1ubuntu1
amd64 Load the NVIDIA kernel driver and create device files
ii nvidia-open-570 570.211.01-0ubuntu1
amd64 NVIDIA Driver meta-package, Open GPU kernel modules, 570 branch-
specific
ii nvidia-persistenced 610.57.04-1ubuntu1
amd64 daemon to maintain persistent software state in the NVIDIA driver
ii nvidia-settings 610.57.04-1ubuntu1
amd64 Tool for configuring the NVIDIA graphics driver
ii screen-resolution-extra 0.18.3ubuntu0.24.04.1
all Extension for the nvidia-settings control panel
ii xserver-xorg-video-nvidia-570 570.211.01-0ubuntu1
amd64 NVIDIA binary Xorg driver
@-h200:~$

```

```
@[REDACTED]-h200:~$
nvidia_uvm          2121728  4
nvidia_drm           131072  0
nvidia_modeset       1728512  1 nvidia_drm
nvidia               11657216 148 nvidia_uvm,nvidia_modeset
video                77824  1 nvidia_modeset
ecc                  45056  1 nvidia
```

```
@[REDACTED]-h200:~$
```

```
@[REDACTED]-h200:~$
nvidia/570.211.01, 6.8.0-138-generic, x86_64: installed
```

```
@[REDACTED]-h200:~$
```

```
[REDACTED]-h200:~$
```

```
• nvidia-fabricmanager.service - NVIDIA fabric manager service
   Loaded: loaded (/usr/lib/systemd/system/nvidia-fabricmanager.service; enabled;
   preset: enabled)
   Active: active (running) since Thu 2026-08-20 13:29:28 UTC; 22min ago
     Process: 4413 ExecStart=/usr/bin/nvidia-fabricmanager-start.sh --mode start
   (code=exited, status=0/SUCCESS)
    Main PID: 4418 (nv-fabricmanager)
       Tasks: 18 (limit: 629145)
      Memory: 22.7M (peak: 24.8M)
         CPU: 547ms
    CGroup: /system.slice/nvidia-fabricmanager.service
            └─4418 /usr/bin/nv-fabricmanager -c
/usr/share/nvidia/nvswitch/fabricmanager.cfg
```

```
Aug 20 13:29:27 [REDACTED]-h200 systemd[1]: Starting nvidia-fabricmanager.service -
NVIDIA fabric manager service...
Aug 20 13:29:27 [REDACTED]-h200 nvidia-fabricmanager-start.sh[4413]: Detected Pre-NVL5
system
Aug 20 13:29:28 [REDACTED]-h200 nv-fabricmanager[4418]: Connected to 1 node.
Aug 20 13:29:28 [REDACTED]-h200 nv-fabricmanager[4418]: Successfully configured all
the available NVSwitches to route GPU NVLink >
Aug 20 13:29:28 [REDACTED]-h200 nvidia-fabricmanager-start.sh[4413]: Started "Nvidia
Fabric Manager"
Aug 20 13:29:28 [REDACTED]-h200 systemd[1]: Started nvidia-fabricmanager.service -
NVIDIA fabric manager service.
```

```
@[REDACTED]-h200:~$
```

```
@[REDACTED]-h200:~$
```

```
• nvidia-dcgm.service - NVIDIA DCGM service
   Loaded: loaded (/usr/lib/systemd/system/nvidia-dcgm.service; enabled; preset:
   enabled)
   Active: active (running) since Thu 2026-08-20 13:29:22 UTC; 22min ago
     Main PID: 4024 (nv-hostengine)
       Tasks: 9 (limit: 629145)
      Memory: 142.5M (peak: 567.0M)
         CPU: 19.388s
    CGroup: /system.slice/nvidia-dcgm.service
            └─4024 /usr/bin/nv-hostengine -n --service-account nvidia-dcgm
```

```
Aug 20 13:29:22 [REDACTED]-h200 systemd[1]: Started nvidia-dcgm.service - NVIDIA DCGM
```


service.

Aug 20 13:29:28 [REDACTED]-h200 nv-hostengine[4024]: DCGM initialized

Aug 20 13:29:28 [REDACTED]-h200 nv-hostengine[4024]: Started host engine version 3.3.9
using port number: 5555

@ [REDACTED]-h200:~\$

@ [REDACTED]-h200:~\$

Thu Aug 20 11:13:31 2026

-----+-----									
NVIDIA-SMI 570.211.01			Driver Version: 570.211.01				CUDA Version:		
12.8									
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----									
+-----+									
GPU Name			Persistence-M		Bus-Id		Disp.A		Volatile
Uncorr. ECC									
Fan Temp Perf			Pwr:Usage/Cap				Memory-Usage		GPU-Util
Compute M.									
MIG M.									
=====+=====+=====+=====+=====+=====+=====+=====+=====+=====									
=====									
0 NVIDIA H200			On		00000000:18:00.0		Off		
0									
N/A 34C P0			74W / 700W		0MiB / 143771MiB				0%
Default									
Disabled									
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----									
+-----+									
1 NVIDIA H200			On		00000000:29:00.0		Off		
0									
N/A 37C P0			75W / 700W		0MiB / 143771MiB				0%
Default									
Disabled									
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----									
+-----+									
2 NVIDIA H200			On		00000000:3A:00.0		Off		
0									
N/A 33C P0			73W / 700W		0MiB / 143771MiB				0%
Default									
Disabled									
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----									
+-----+									
3 NVIDIA H200			On		00000000:4B:00.0		Off		
0									
N/A 56C P0			85W / 700W		0MiB / 143771MiB				0%
Default									
Disabled									
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----									
+-----+									
4 NVIDIA H200			On		00000000:9A:00.0		Off		
0									
N/A 34C P0			75W / 700W		0MiB / 143771MiB				0%

```

Default |
|
Disabled |
+-----+-----+
| 5 NVIDIA H200 On | 00000000:AA:00.0 Off |
0 |
| N/A 36C P0 76W / 700W | 0MiB / 143771MiB | 0%
Default |
|
Disabled |
+-----+-----+
| 6 NVIDIA H200 On | 00000000:BA:00.0 Off |
0 |
| N/A 35C P0 75W / 700W | 0MiB / 143771MiB | 0%
Default |
|
Disabled |
+-----+-----+
| 7 NVIDIA H200 On | 00000000:CA:00.0 Off |
0 |
| N/A 55C P0 83W / 700W | 0MiB / 143771MiB | 0%
Default |
|
Disabled |
+-----+-----+

```

```

+-----+
| Processes:
| GPU GI CI PID Type Process name GPU
Memory |
| ID ID
Usage |
=====
=====|
| No running processes found
|
+-----+

```

@[REDACTED]-h200:~\$

@[REDACTED]-h200:~\$

	GPU0	GPU1	GPU2	GPU3	GPU4	GPU5	GPU6	GPU7	CPU
Affinity									
GPU0	X	NV18	NV18	NV18	NV18	NV18	NV18	NV18	0-47,96-143
0		N/A							
GPU1	NV18	X	NV18	NV18	NV18	NV18	NV18	NV18	0-47,96-143
0		N/A							
GPU2	NV18	NV18	X	NV18	NV18	NV18	NV18	NV18	0-47,96-143
0		N/A							
GPU3	NV18	NV18	NV18	X	NV18	NV18	NV18	NV18	0-47,96-143
0		N/A							

GPU4	NV18	NV18	NV18	NV18	X	NV18	NV18	NV18	48-95,144-
191	1		N/A						
GPU5	NV18	NV18	NV18	NV18	NV18	X	NV18	NV18	48-95,144-
191	1		N/A						
GPU6	NV18	NV18	NV18	NV18	NV18	NV18	X	NV18	48-95,144-
191	1		N/A						
GPU7	NV18	NV18	NV18	NV18	NV18	NV18	NV18	X	48-95,144-
191	1		N/A						

Legend:

X = Self
 SYS = Connection traversing PCIe as well as the SMP interconnect between NUMA nodes (e.g., QPI/UPI)
 NODE = Connection traversing PCIe as well as the interconnect between PCIe Host Bridges within a NUMA node
 PHB = Connection traversing PCIe as well as a PCIe Host Bridge (typically the CPU)
 PXB = Connection traversing multiple PCIe bridges (without traversing the PCIe Host Bridge)
 PIX = Connection traversing at most a single PCIe bridge
 NV# = Connection traversing a bonded set of # NVLinks

@llmus02-h200:~\$

@[REDACTED]-h200:~\$

8 GPUs found.

GPU ID	Device Information
0	Name: NVIDIA H200 PCI Bus ID: 00000000:18:00.0 Device UUID: GPU-ae42bc94-1b65-98d6-0eb5-c3e6efa0c24c
1	Name: NVIDIA H200 PCI Bus ID: 00000000:29:00.0 Device UUID: GPU-d31aee23-abbd-3704-1b50-c4993121ea74
2	Name: NVIDIA H200 PCI Bus ID: 00000000:3A:00.0 Device UUID: GPU-4f46484c-b1c3-f791-21cd-4ce3f9cea953
3	Name: NVIDIA H200 PCI Bus ID: 00000000:4B:00.0 Device UUID: GPU-a891943e-b3b9-bef6-8495-18b7da7b68ad
4	Name: NVIDIA H200 PCI Bus ID: 00000000:9A:00.0 Device UUID: GPU-2b5f9c65-18f9-f74e-2f23-3435206ac9fc
5	Name: NVIDIA H200 PCI Bus ID: 00000000:AA:00.0 Device UUID: GPU-b1f2d02a-4723-47e4-c3af-c8fea63c69b3
6	Name: NVIDIA H200 PCI Bus ID: 00000000:BA:00.0 Device UUID: GPU-953b3930-a29f-6b39-719a-3943726f6912
7	Name: NVIDIA H200

```

|          | PCI Bus ID: 00000000:CA:00.0          |
|          | Device UUID: GPU-6ef99506-a8c3-99a2-a466-1fe631c93a3c |
+-----+
4 NVSwitches found.
+-----+
| Switch ID |
+-----+
| 2          |
| 0          |
| 1          |
| 3          |
+-----+
0 CPUs found.
+-----+
| CPU ID | Device Information |
+-----+
+-----+

```

██████-h200:~\$

```

+-----+
+-----+
| Health Monitor Report |
|                         |
+=====+
====+
| Overall Health      | Healthy |
|                     |         |
+-----+
+-----+

```

██████-h200:~\$

```

██████-h200:~$
nvcc: NVIDIA (R) Cuda compiler driver
Copyright (c) 2005-2026 NVIDIA Corporation
Built on Tue_Jun_09_02:43:40_PM_PDT_2026
Cuda compilation tools, release 13.3, V13.3.73
Build cuda_13.3.r13.3/compiler.38244171_0
██████-h200:~$

```

██████-h200:~\$

```

Successfully ran diagnostic for group.
+-----+
| Diagnostic | Result |
+-----+
+-----+
| ----- Metadata ----- |
| DCGM Version | 3.3.9 |
| Driver Version Detected | 570.211.01 |
| GPU Device IDs Detected | 2335, 2335, 2335, 2335, 2335, 2335, 2335, 2335 |
| ----- Deployment ----- |
| Denylist | Pass |
| NVML Library | Pass |
| CUDA Main Library | Pass |
| Permissions and OS Blocks | Pass |
| Persistence Mode | Pass |
| Environment Variables | Pass |
| Page Retirement/Row Remap | Pass |
+-----+

```

Graphics Processes	Pass
Inforom	Pass

@-h200:~\$

@-h200:~\$

Successfully ran diagnostic for group.

Diagnostic	Result
----- Metadata -----	
DCGM Version	3.3.9
Driver Version Detected	570.211.01
GPU Device IDs Detected	2335, 2335, 2335, 2335, 2335, 2335, 2335, 2335
----- Deployment -----	
Denylist	Pass
NVML Library	Pass
CUDA Main Library	Pass
Permissions and OS Blocks	Pass
Persistence Mode	Pass
Environment Variables	Pass
Page Retirement/Row Remap	Pass
Graphics Processes	Pass
Inforom	Pass
----- Integration -----	
PCIe	Pass - GPUs: 0, 1, 2, 4, 5, 6, 7 Fail - GPU: 3
Warning	GPU 3 There were thermal violations totaling 8548.2 seconds for GPU 3 Verify that the cooling on this machine is functional, including external, thermal material interface, fans, and any other components.
Info	GPU 0 GPU to Host bandwidth: 51.49 GB/s, GPU 0 Host to GPU bandwidth: 54.01 GB/s, GPU 0 bidirectional bandwidth: 92.62 GB/s, GPU 0 GPU to Host latency: 1.937 us, GPU 0 Host to GPU latency: 2.161 us, GPU 0 bidirectional latency: 3.312 us
Info	GPU 1 GPU to Host bandwidth: 51.49 GB/s, GPU 1 Host to GPU bandwidth: 54.03 GB/s, GPU 1 bidirectional bandwidth: 93.59 GB/s, GPU 1 GPU to Host latency: 1.926 us, GPU 1 Host to GPU latency: 2.143 us, GPU 1 bidirectional latency: 3.507 us
Info	GPU 2 GPU to Host bandwidth: 51.49 GB/s, GPU 2 Host to GPU bandwidth: 54.06 GB/s, GPU 2 bidirectional bandwidth: 93.67 GB/s, GPU 2 GPU to Host latency: 1.947 us, GPU 2 Host to GPU latency: 2.161 us, GPU 2 bidirectional latency: 3.760 us
Info	GPU 3 GPU to Host bandwidth: 51.49 GB/s, GPU 3 Host to GPU bandwidth: 54.05 GB/s, GPU 3 bidirectional bandwidth: 93.55 GB/s, GPU 3 GPU to Host latency: 1.982 us, GPU 3 Host to GPU latency: 2.237 us, GPU 3 bidirectional latency: 3.957 us
Info	GPU 4 GPU to Host bandwidth: 51.49 GB/s, GPU 4 Host to GPU bandwidth: 54.00 GB/s, GPU 4

	bidirectional bandwidth: 92.36 GB/s, GPU 4 GPU to Host latency: 2.193 us, GPU 4 Host to GPU latency: 2.600 us, GPU 4 bidirectional latency: 4.686 us
Info	GPU 5 GPU to Host bandwidth: 51.48 GB/s, GPU 5 Host to GPU bandwidth: 54.00 GB/s, GPU 5 bidirectional bandwidth: 93.62 GB/s, GPU 5 GPU to Host latency: 2.392 us, GPU 5 Host to GPU latency: 2.463 us, GPU 5 bidirectional latency: 4.617 us
Info	GPU 6 GPU to Host bandwidth: 51.50 GB/s, GPU 6 Host to GPU bandwidth: 54.01 GB/s, GPU 6 bidirectional bandwidth: 74.08 GB/s, GPU 6 GPU to Host latency: 2.484 us, GPU 6 Host to GPU latency: 2.442 us, GPU 6 bidirectional latency: 4.809 us
Info	GPU 7 GPU to Host bandwidth: 51.50 GB/s, GPU 7 Host to GPU bandwidth: 54.01 GB/s, GPU 7 bidirectional bandwidth: 93.72 GB/s, GPU 7 GPU to Host latency: 2.430 us, GPU 7 Host to GPU latency: 2.395 us, GPU 7 bidirectional latency: 4.708 us
+----- Hardware -----+	
GPU Memory	Pass - GPUs: 0, 1, 2, 4, 5, 6, 7 Fail - GPU: 3
Warning	GPU 3 Thermal violations totaling 14247.0 seconds started at 5.0 seconds into the test for GPU 3 Verify that the cooling on this machine is functional, including external, thermal material interface, fans, and any other components.
Info	GPU 0 Allocated 148068455535 bytes (98.6%)
Info	GPU 1 Allocated 148068455535 bytes (98.6%)
Info	GPU 2 Allocated 148068455535 bytes (98.6%)
Info	GPU 3 Allocated 148068455535 bytes (98.6%)
Info	GPU 4 Allocated 148068455535 bytes (98.6%)
Info	GPU 5 Allocated 148068455535 bytes (98.6%)
Info	GPU 6 Allocated 148068455535 bytes (98.6%)
Info	GPU 7 Allocated 148068455535 bytes (98.6%)
+----- Stress -----+	

h200:~\$

// diagnostics run level 4 fails to complete due to environment constraints

Update



Successfully powered down the system.



Device Name	New Version	Installed Version	Status
NVIDIA H200	96.00.da.00.11	96.00.da.00.11	✓ Updated. OS reboot required.
NVIDIA H200	96.00.da.00.11	96.00.da.00.11	✓ Updated. OS reboot required.
NVIDIA H200	96.00.da.00.11	96.00.da.00.11	✓ Updated. OS reboot required.
NVIDIA H200	96.00.da.00.11	96.00.da.00.11	✓ Updated. OS reboot required.
NVIDIA H200	96.00.da.00.11	96.00.da.00.11	✓ Updated. OS reboot required.
NVIDIA H200	96.00.da.00.11	96.00.da.00.11	✓ Updated. OS reboot required.
NVIDIA H200	96.00.da.00.11	96.00.da.00.11	✓ Updated. OS reboot required.
NVIDIA H200	96.00.da.00.11	96.00.da.00.11	✓ Updated. OS reboot required.

Finish